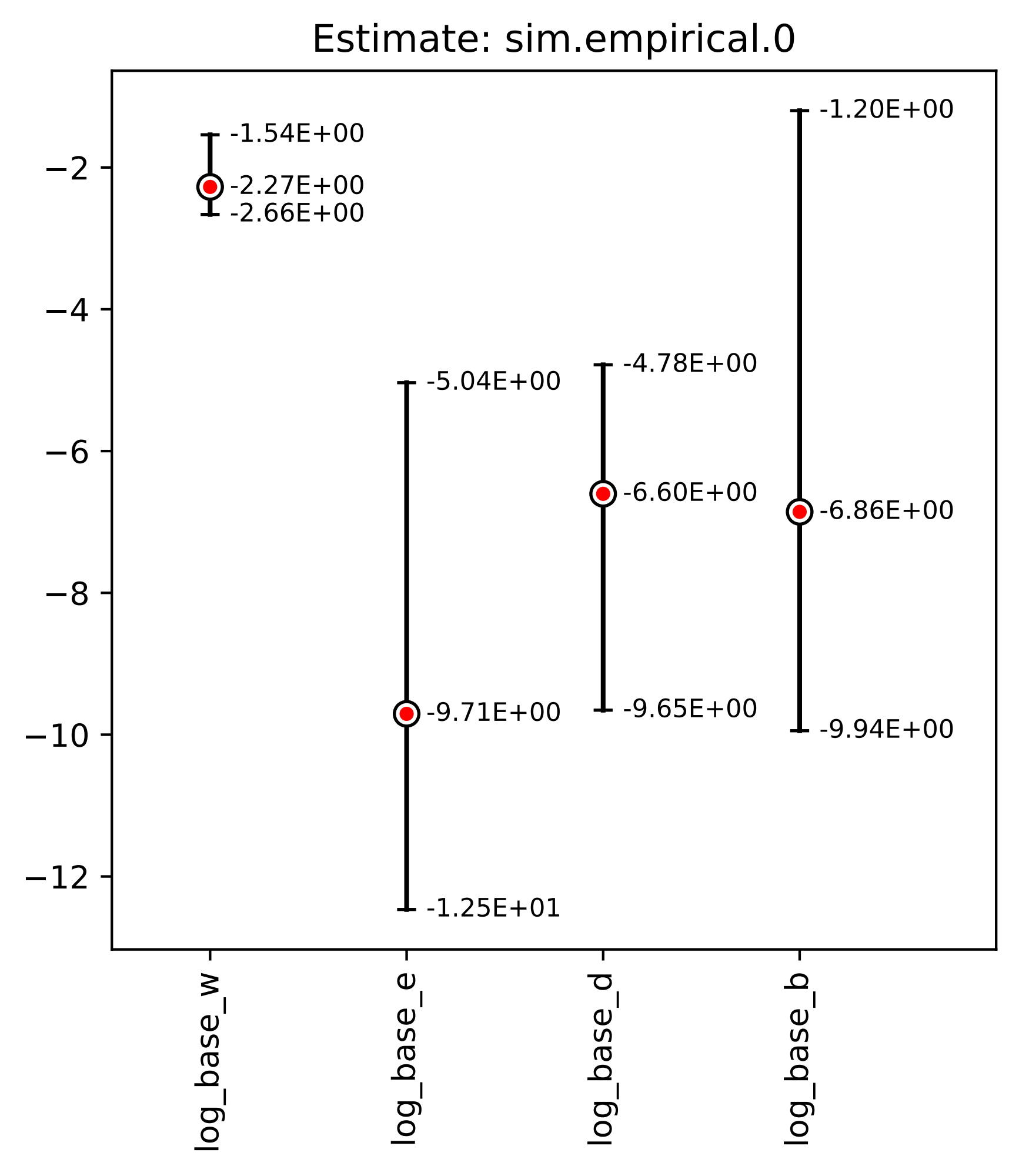
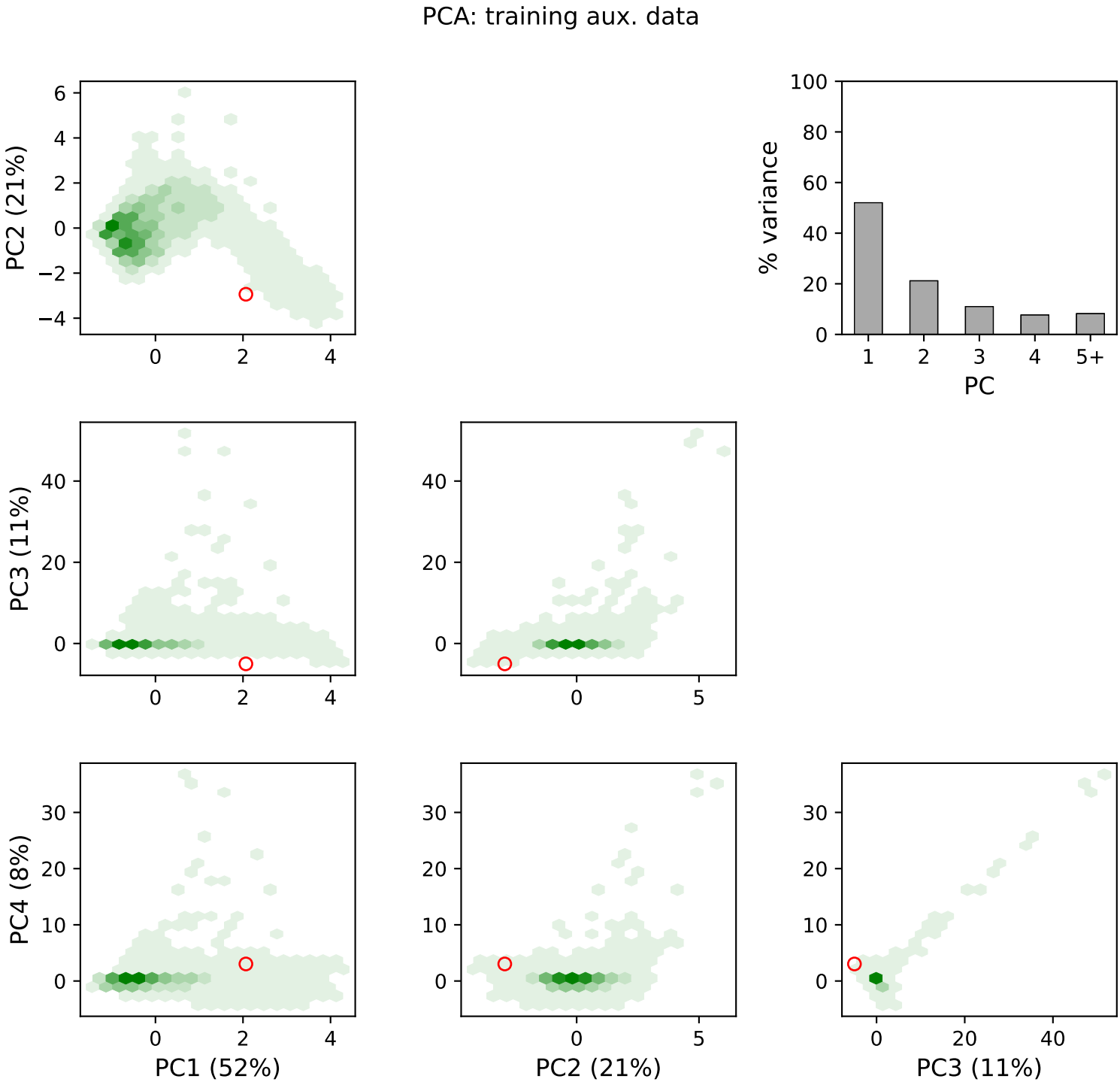
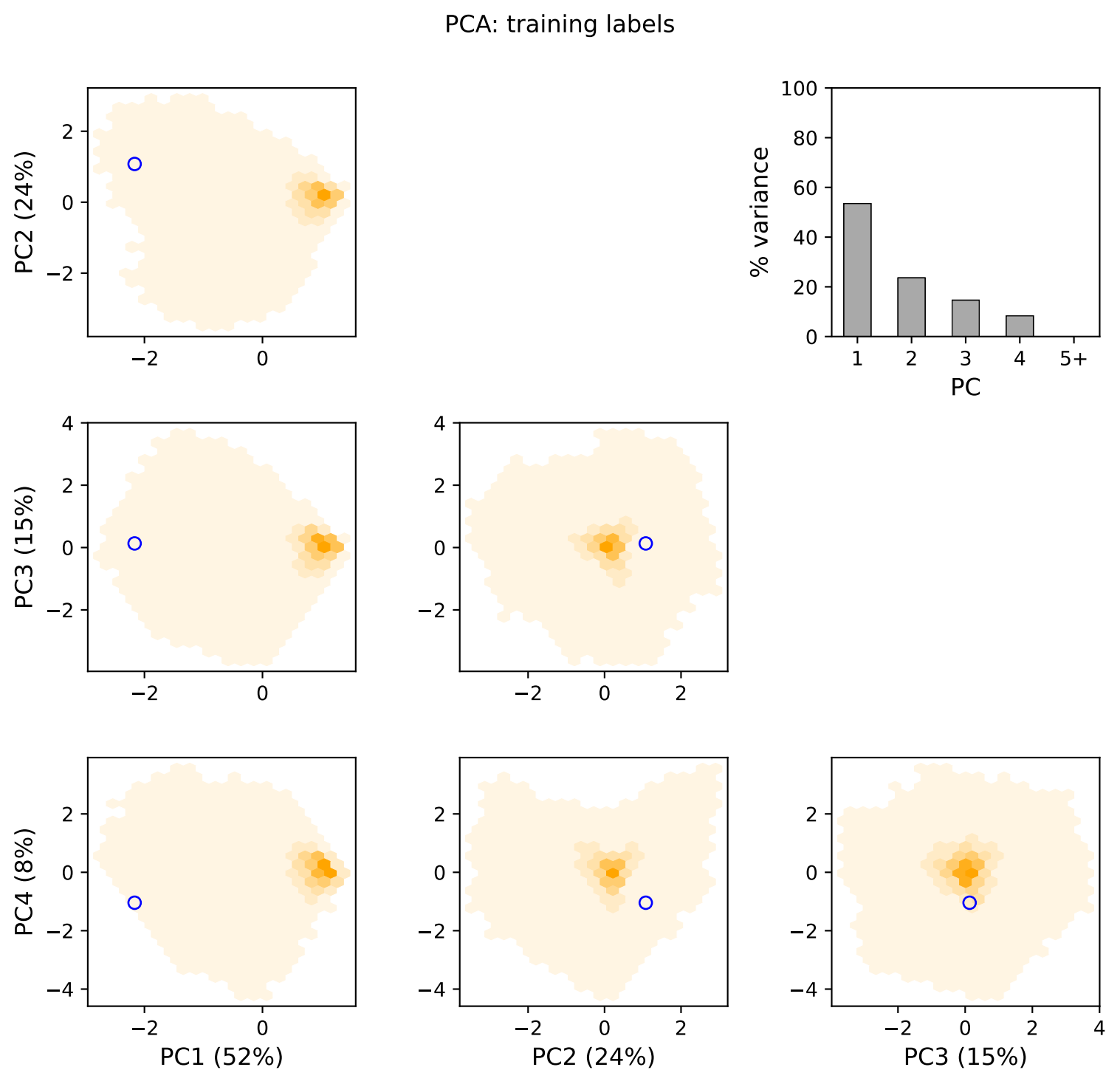
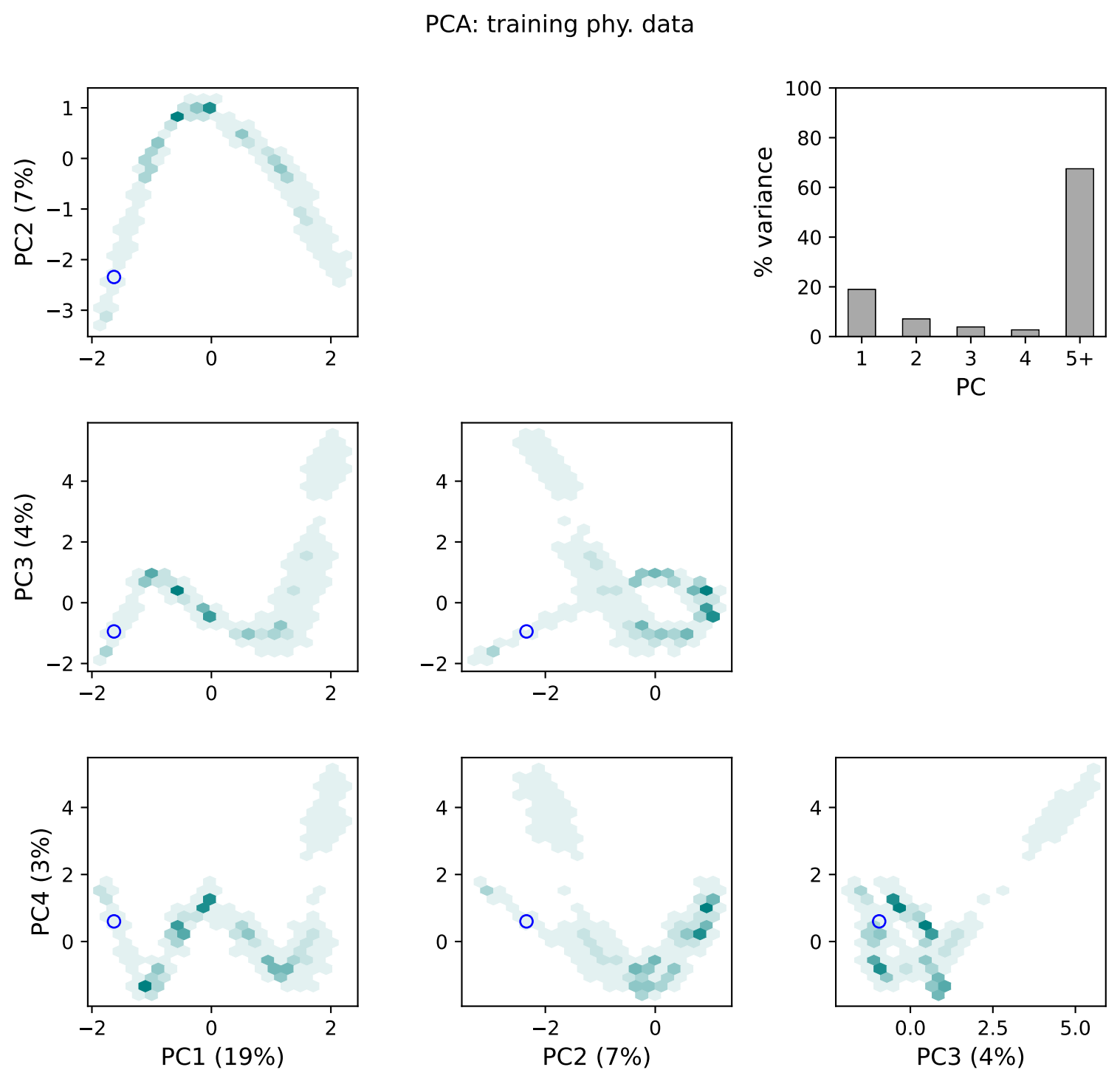
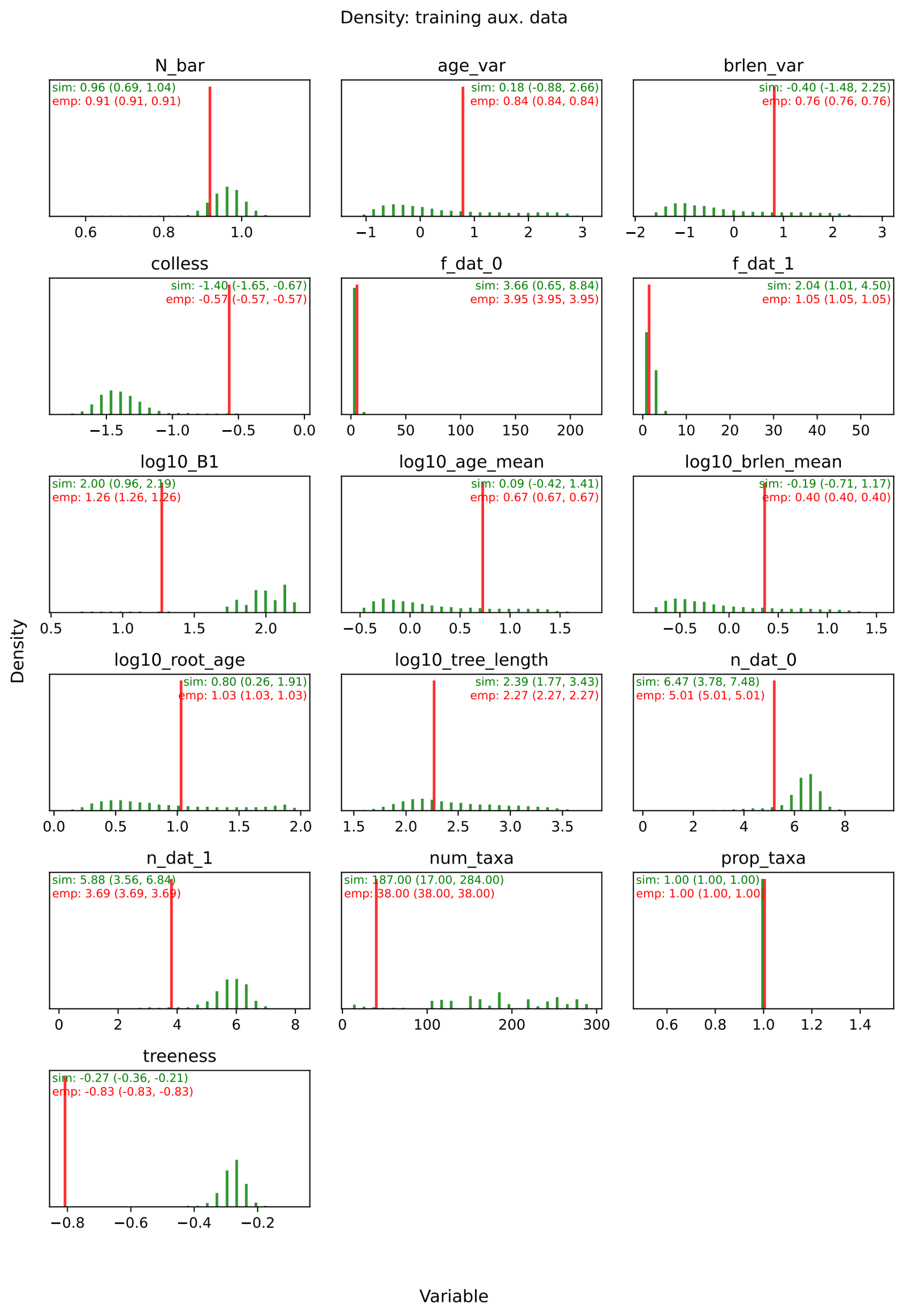
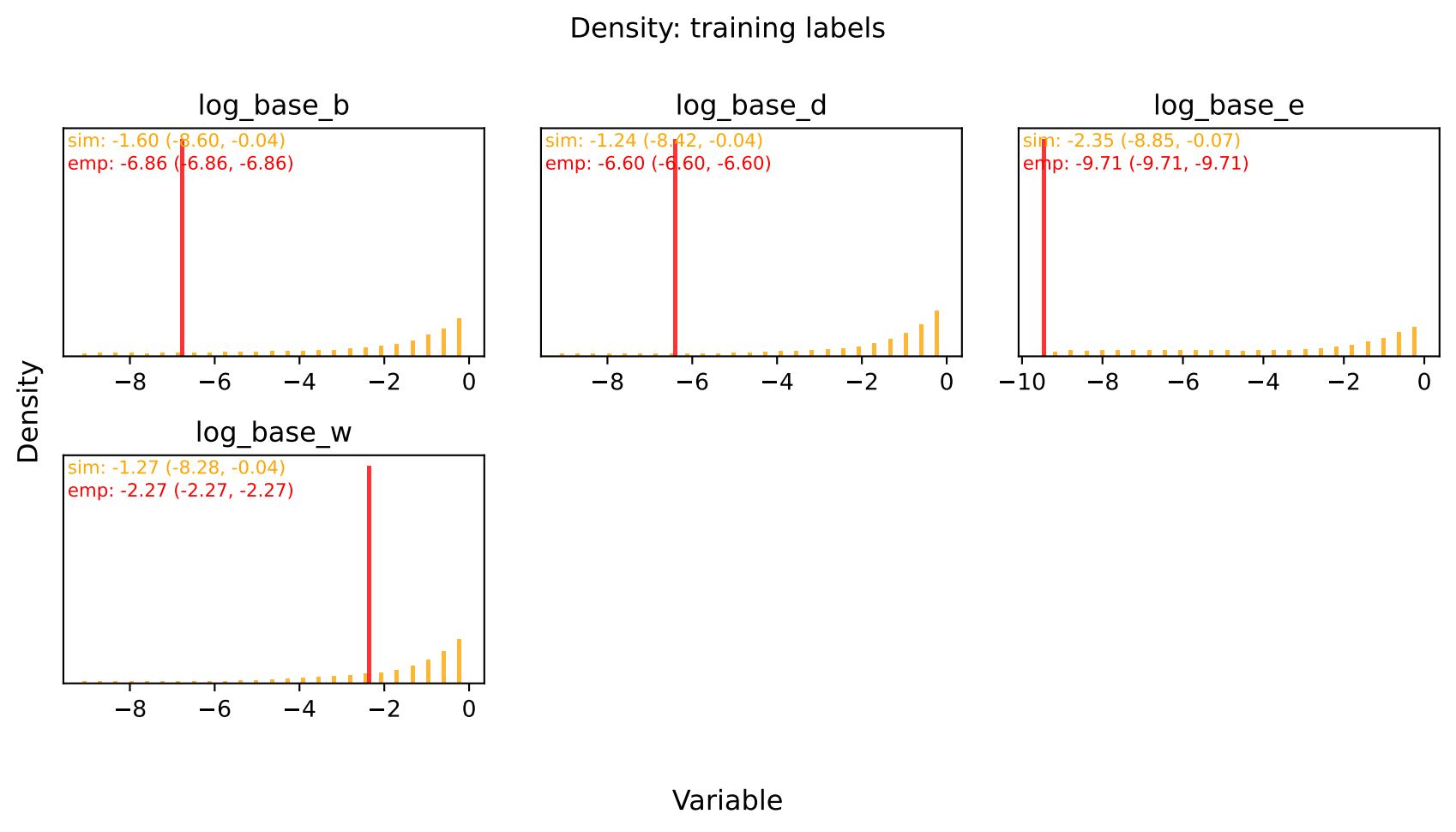

Estimate: sim.empirical.0
−2
−4
−6
−8
−10
−12
-1.54E+00
-2.27E+00
-2.66E+00
-5.04E+00
-9.71E+00
-1.25E+01
-4.78E+00
-6.60E+00
-9.65E+00
-1.20E+00
-6.86E+00
-9.94E+00
log_base_w
log_base_e
log_base_d
log_base_b

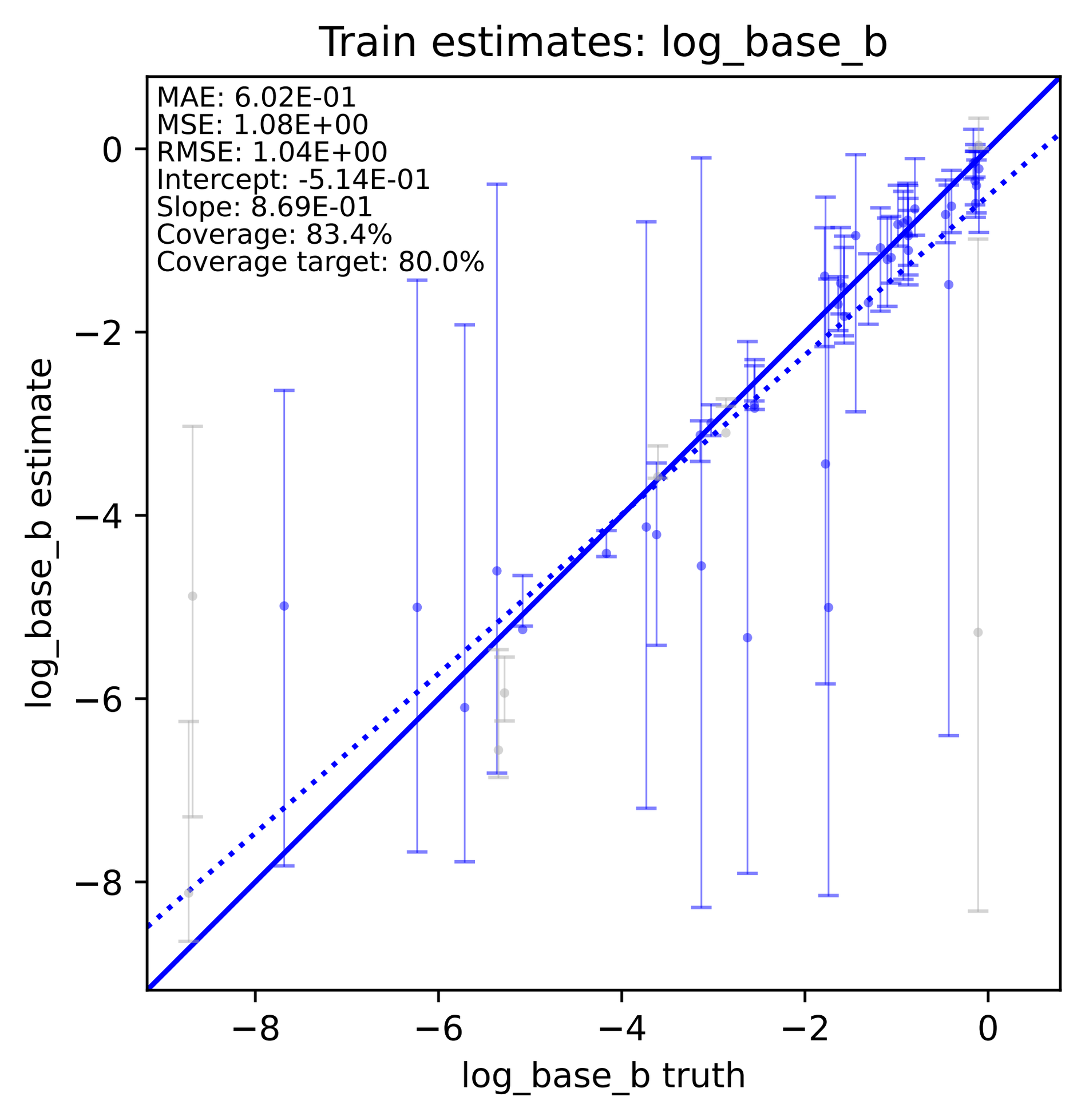

PCA: training aux. data
PC2 (21%)
% variance
PC
PC3 (11%)
PC4 (8%)
PC1 (52%)
PC2 (21%)
PC3 (11%)

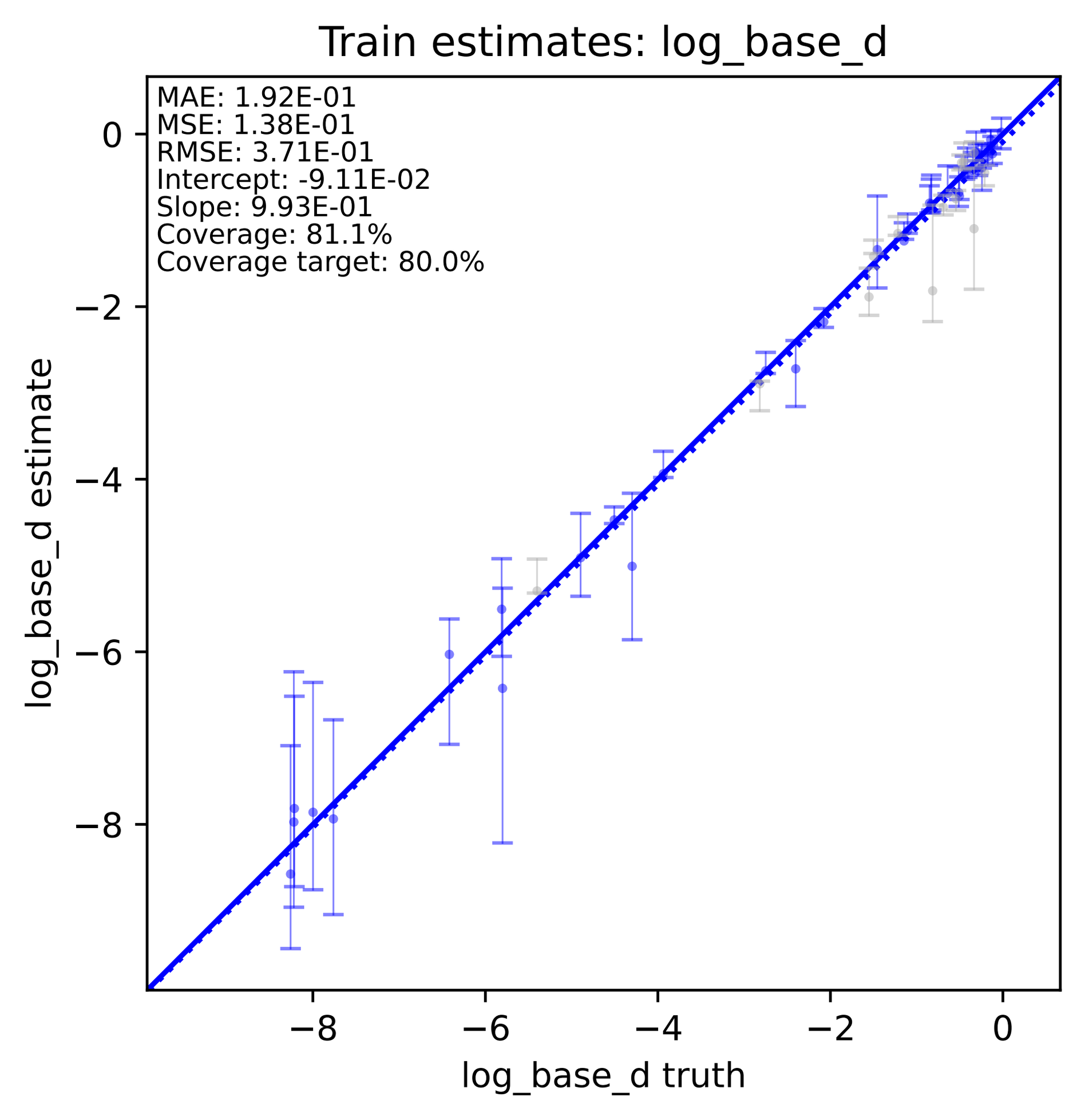

PCA: training labels
PC2 (24%)
% variance
PC
PC3 (15%)
PC4 (8%)
PC1 (52%)
PC2 (24%)
PC3 (15%)

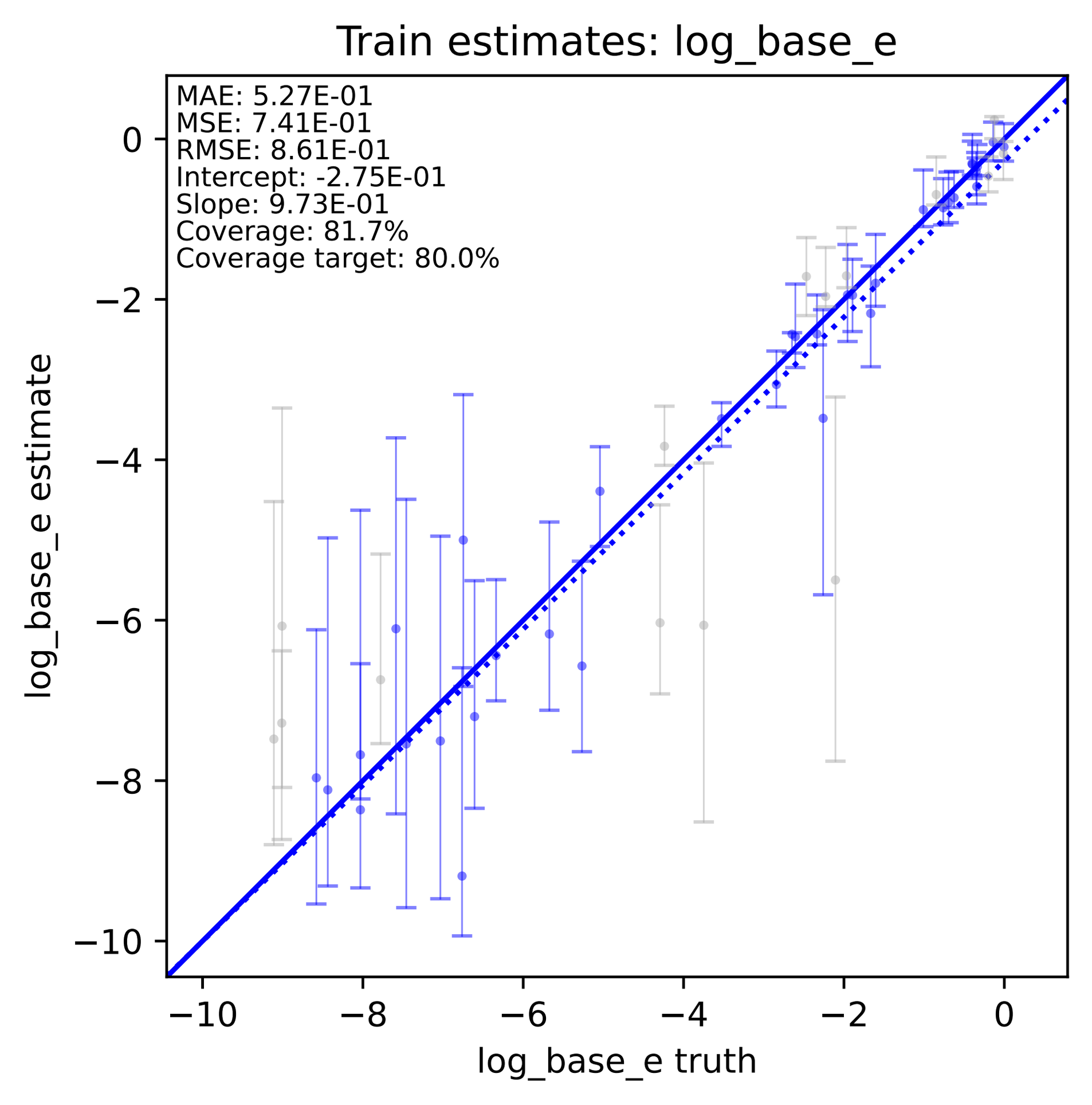

PCA: training phy. data
PC2 (7%)
% variance
PC
PC3 (4%)
PC4 (3%)
PC1 (19%)
PC2 (7%)
PC3 (4%)

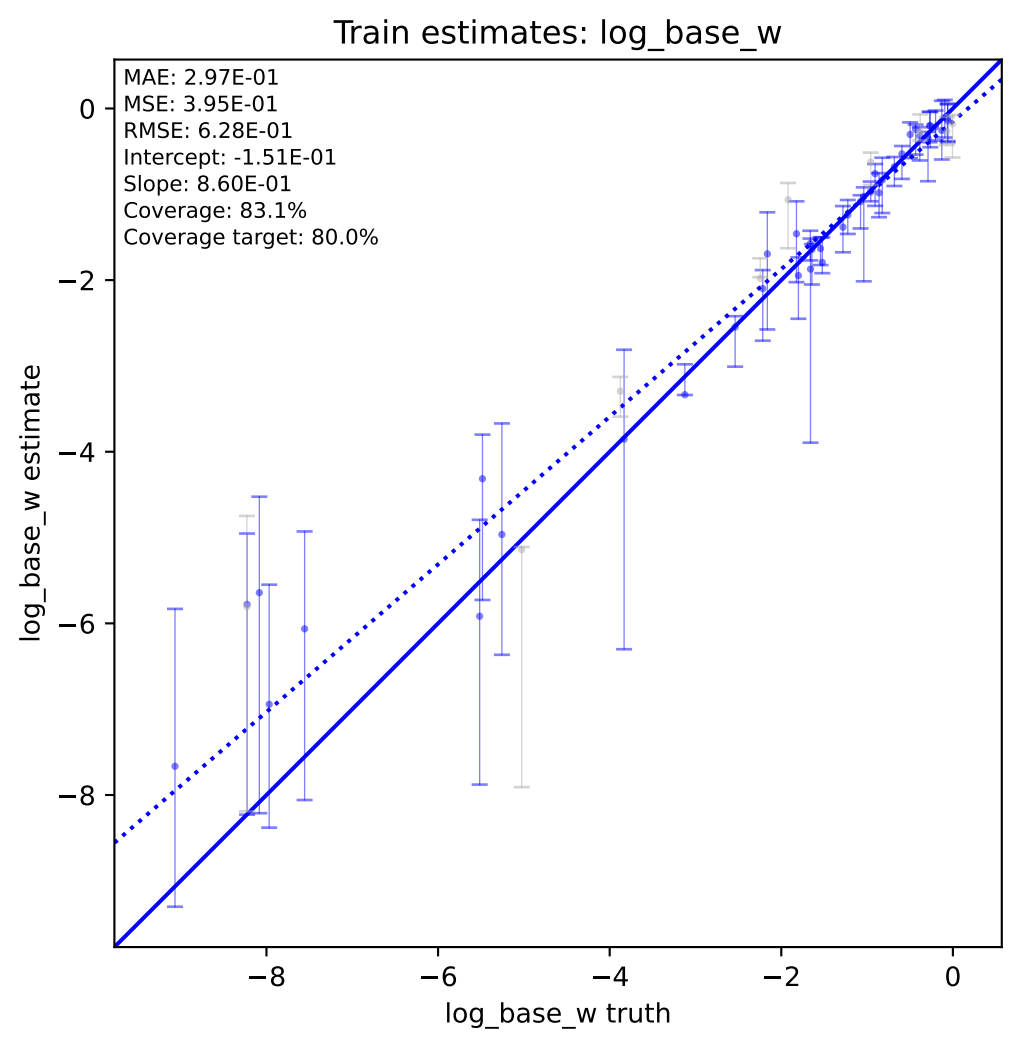

Density: training aux. data
N_bar
sim: 0.96 (0.69, 1.04)
emp: 0.91 (0.91, 0.91)
age_var
sim: 0.18 (-0.88, 2.66)
emp: 0.84 (0.84, 0.84)
brlen_var
sim: -0.40 (-1.48, 2.25)
emp: 0.76 (0.76, 0.76)
colless
sim: -1.40 (-1.65, -0.67)
emp: -0.57 (-0.57, -0.57)
f_dat_0
sim: 3.66 (0.65, 8.84)
emp: 3.95 (3.95, 3.95)
f_dat_1
sim: 2.04 (1.01, 4.50)
emp: 1.05 (1.05, 1.05)
log10_B1
sim: 2.00 (0.96, 2.19)
emp: 1.26 (1.26, 1.26)
log10_age_mean
sim: 0.09 (-0.42, 1.41)
emp: 0.67 (0.67, 0.67)
log10_brlen_mean
sim: -0.19 (-0.71, 1.17)
emp: 0.40 (0.40, 0.40)
log10_root_age
sim: 0.80 (0.26, 1.91)
emp: 1.03 (1.03, 1.03)
log10_tree_length
sim: 2.39 (1.77, 3.43)
emp: 2.27 (2.27, 2.27)
n_dat_0
sim: 6.47 (3.78, 7.48)
emp: 5.01 (5.01, 5.01)
n_dat_1
sim: 5.88 (3.56, 6.84)
emp: 3.69 (3.69, 3.69)
num_taxa
sim: 187.00 (17.00, 284.00)
emp: 38.00 (38.00, 38.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.27 (-0.36, -0.21)
emp: -0.83 (-0.83, -0.83)
Density
Variable

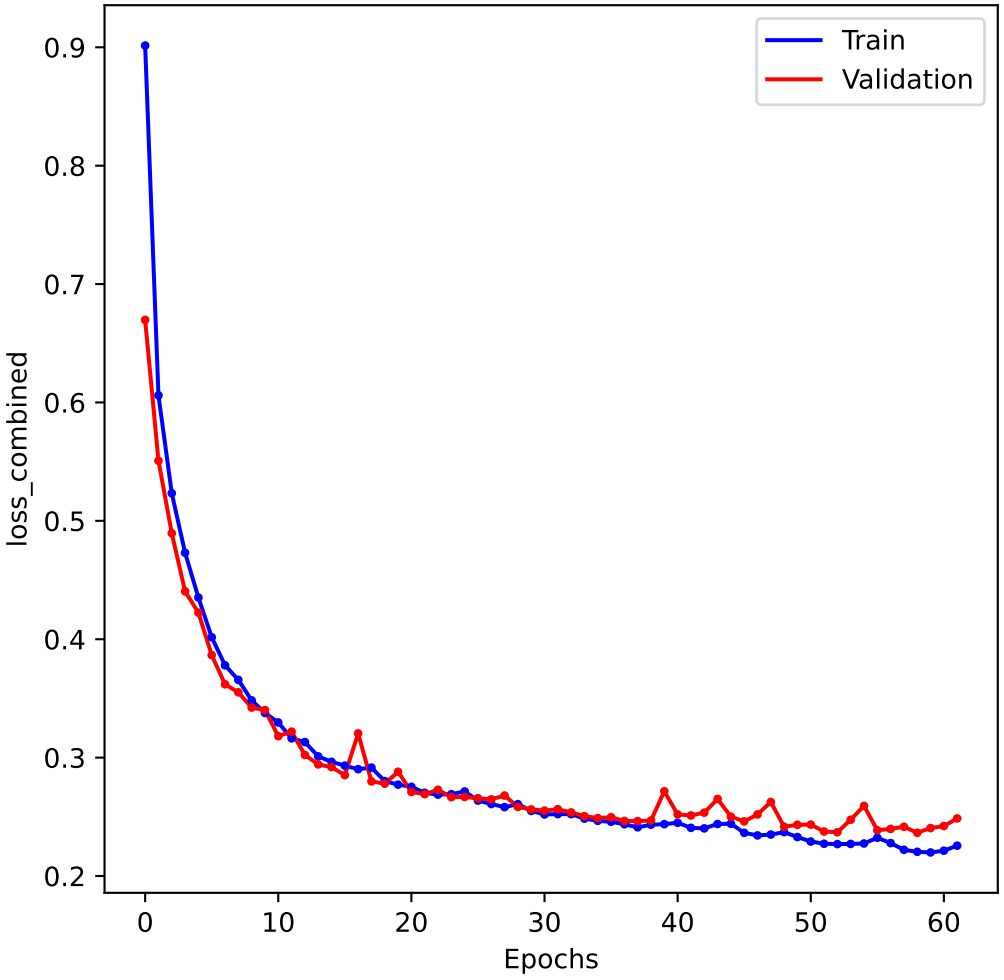

Density: training labels
log_base_b
sim: -1.60 (-8.60, -0.04)
emp: -6.86 (-6.86, -6.86)
log_base_d
sim: -1.24 (-8.42, -0.04)
emp: -6.60 (-6.60, -6.60)
log_base_e
sim: -2.35 (-8.85, -0.07)
emp: -9.71 (-9.71, -9.71)
log_base_w
sim: -1.27 (-8.28, -0.04)
emp: -2.27 (-2.27, -2.27)
Density
Variable

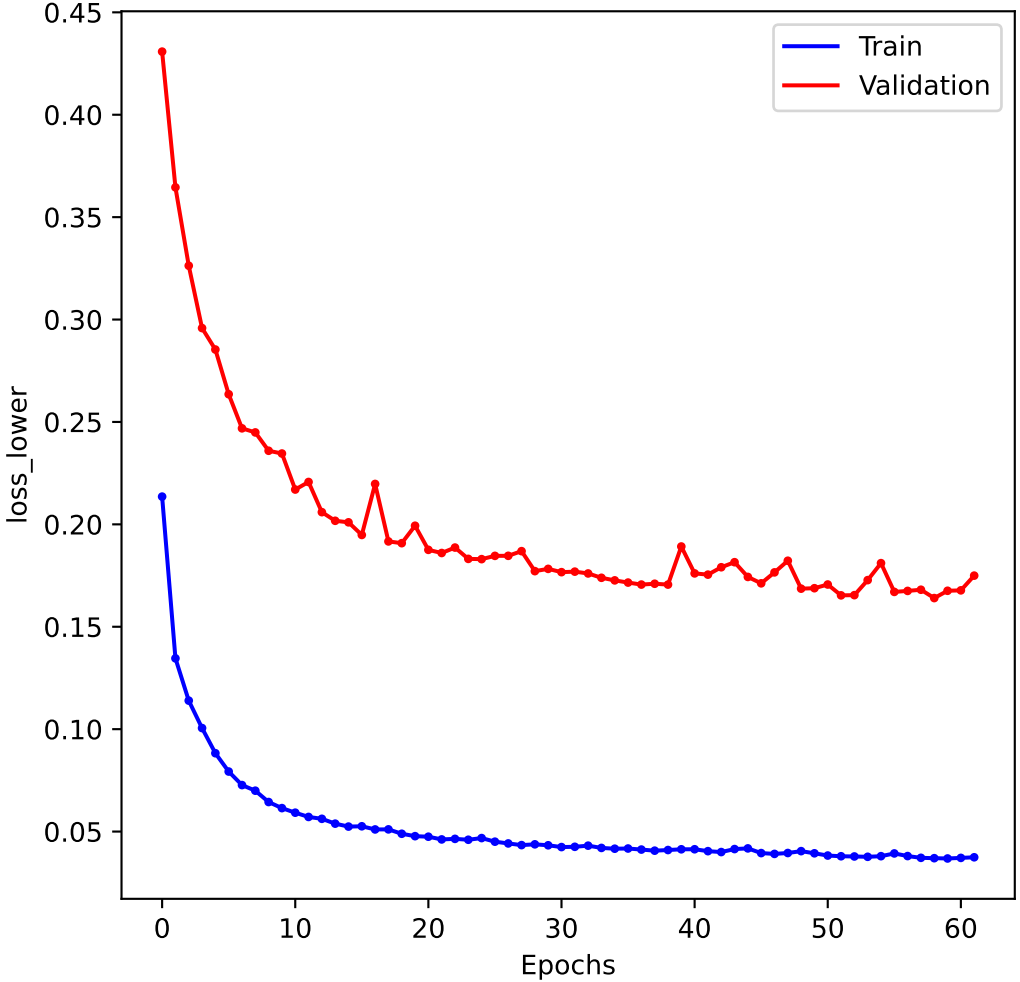

Train estimates: log_base_b
MAE: 6.02E-01
MSE: 1.08E+00
RMSE: 1.04E+00
Intercept: -5.14E-01
Slope: 8.69E-01
Coverage: 83.4%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

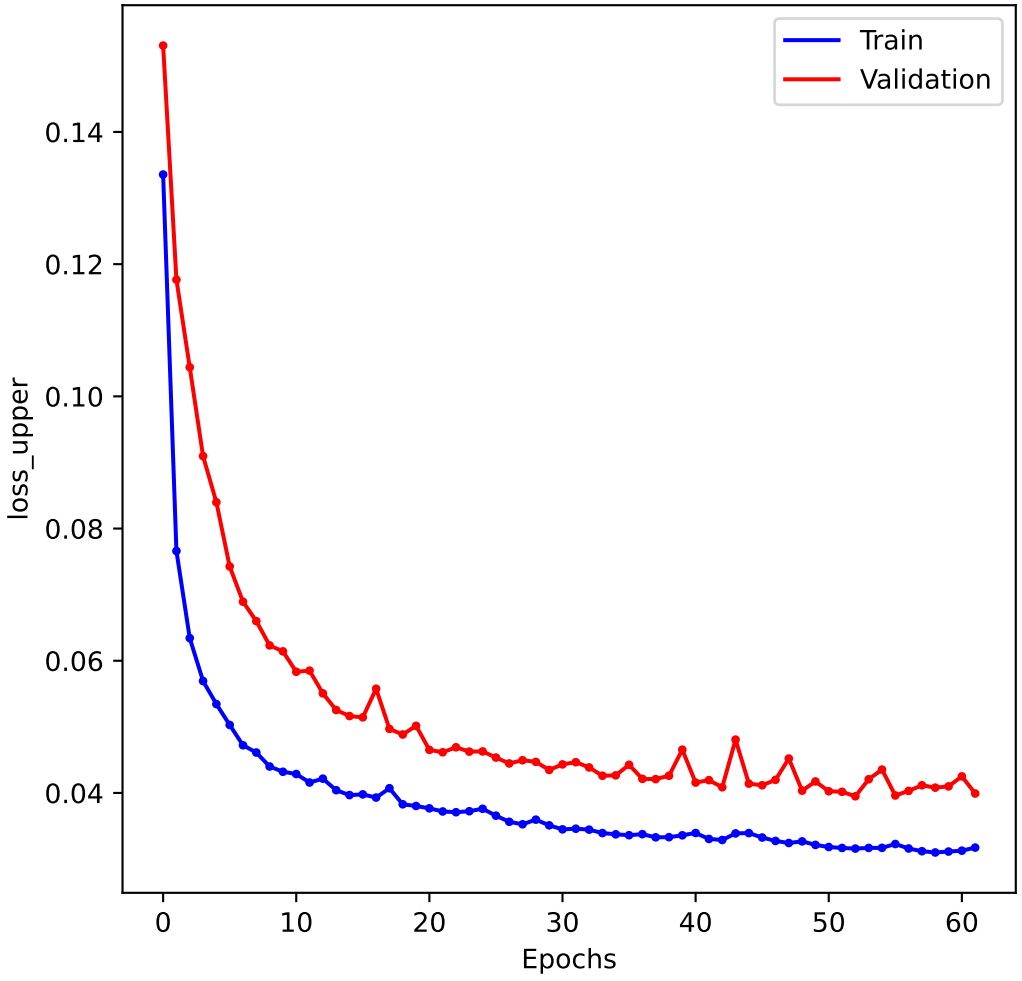

Train estimates: log_base_d
MAE: 1.92E-01
MSE: 1.38E-01
RMSE: 3.71E-01
Intercept: -9.11E-02
Slope: 9.93E-01
Coverage: 81.1%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

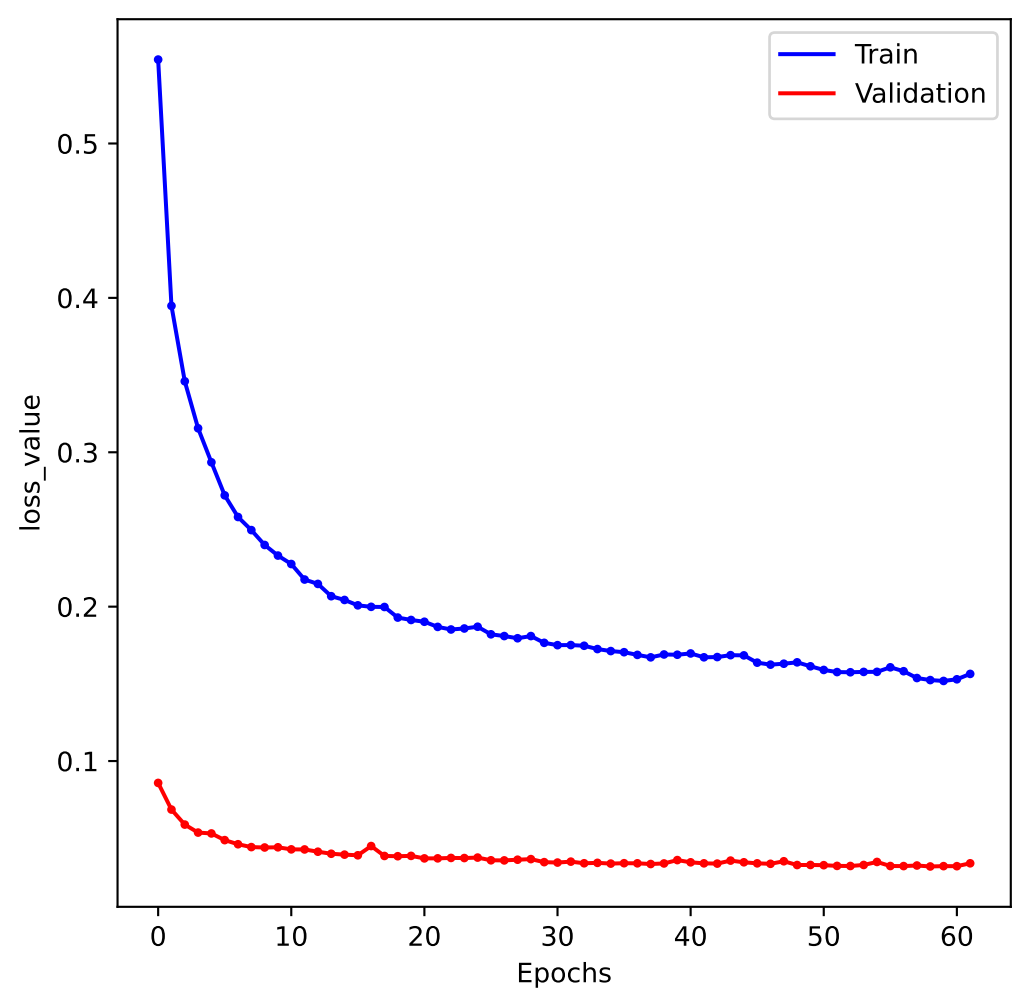

Train estimates: log_base_e
MAE: 5.27E-01
MSE: 7.41E-01
RMSE: 8.61E-01
Intercept: -2.75E-01
Slope: 9.73E-01
Coverage: 81.7%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

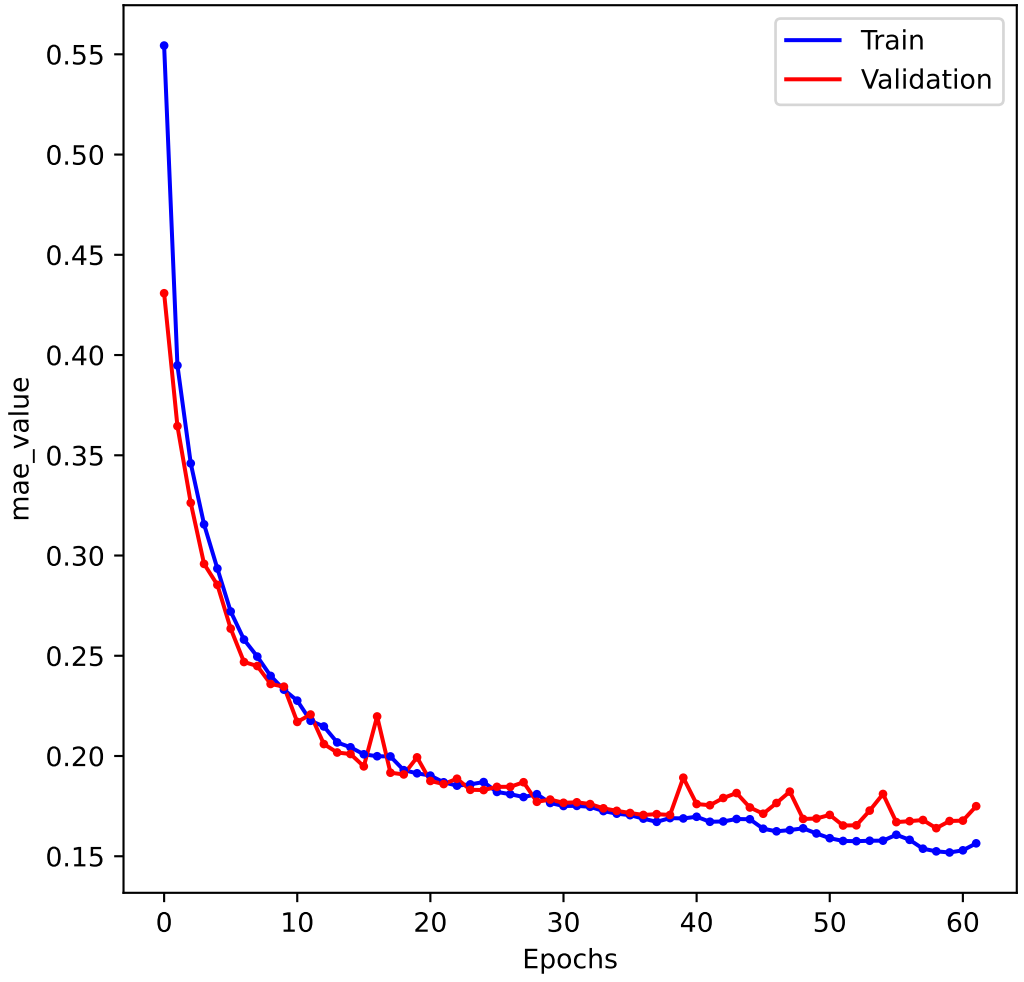

Train estimates: log_base_w
MAE: 2.97E-01
MSE: 3.95E-01
RMSE: 6.28E-01
Intercept: -1.51E-01
Slope: 8.60E-01
Coverage: 83.1%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

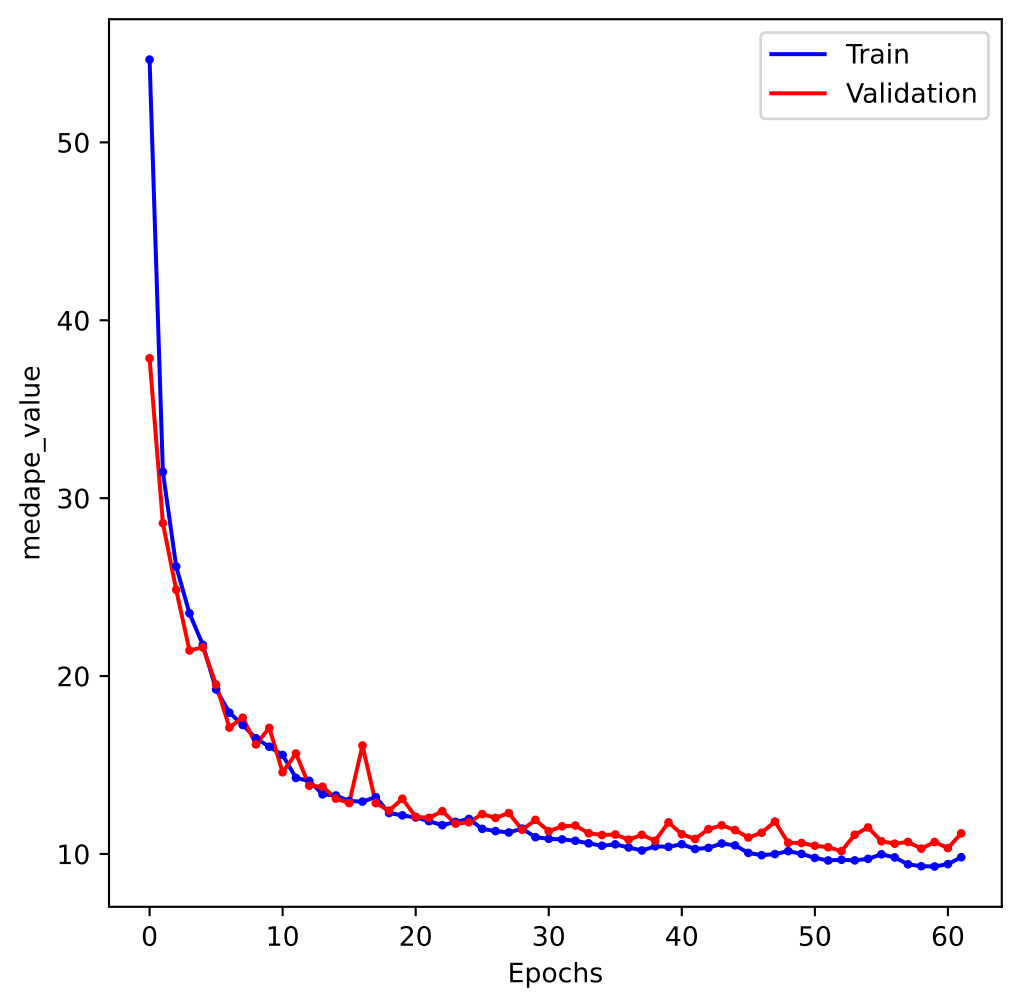

Train
Validation
loss_combined
Epochs
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
10
20
30
40
50
60

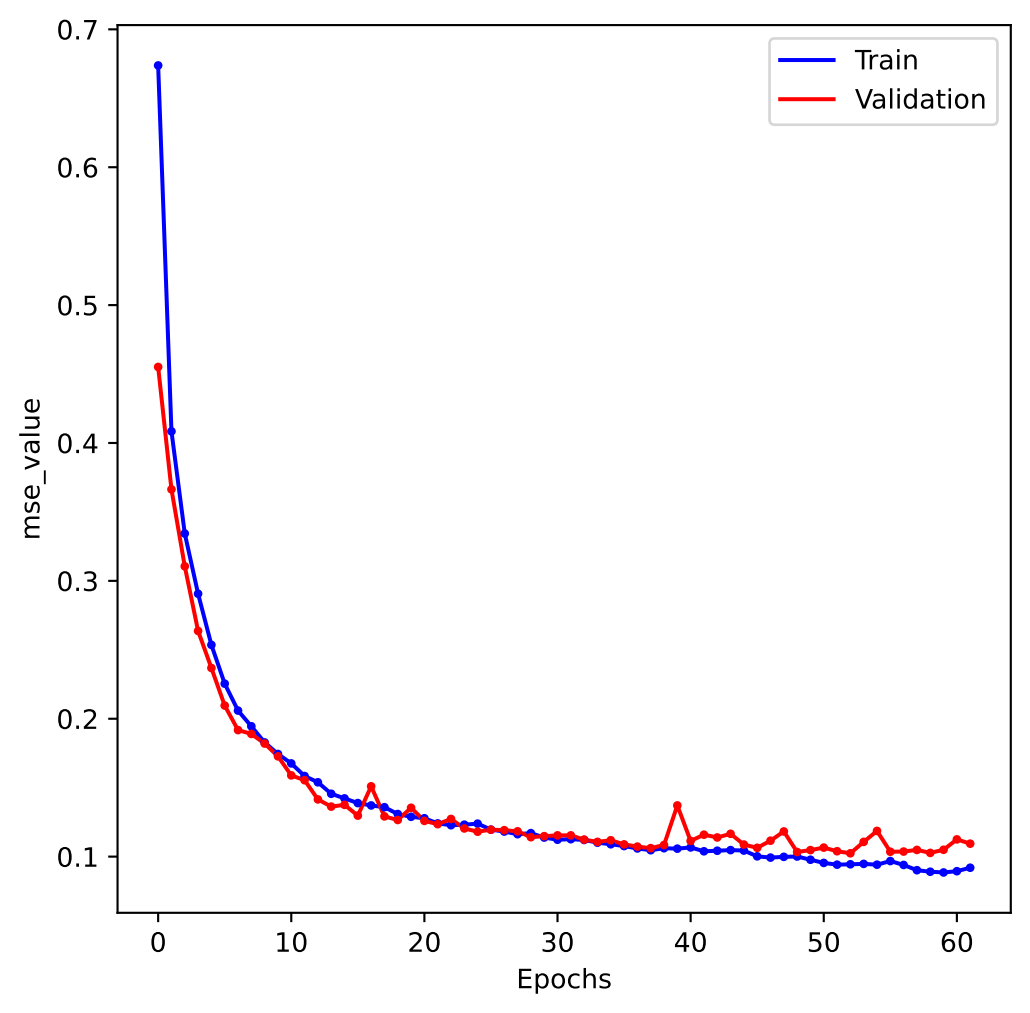

Train
Validation
loss_lower
Epochs
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0
10
20
30
40
50
60

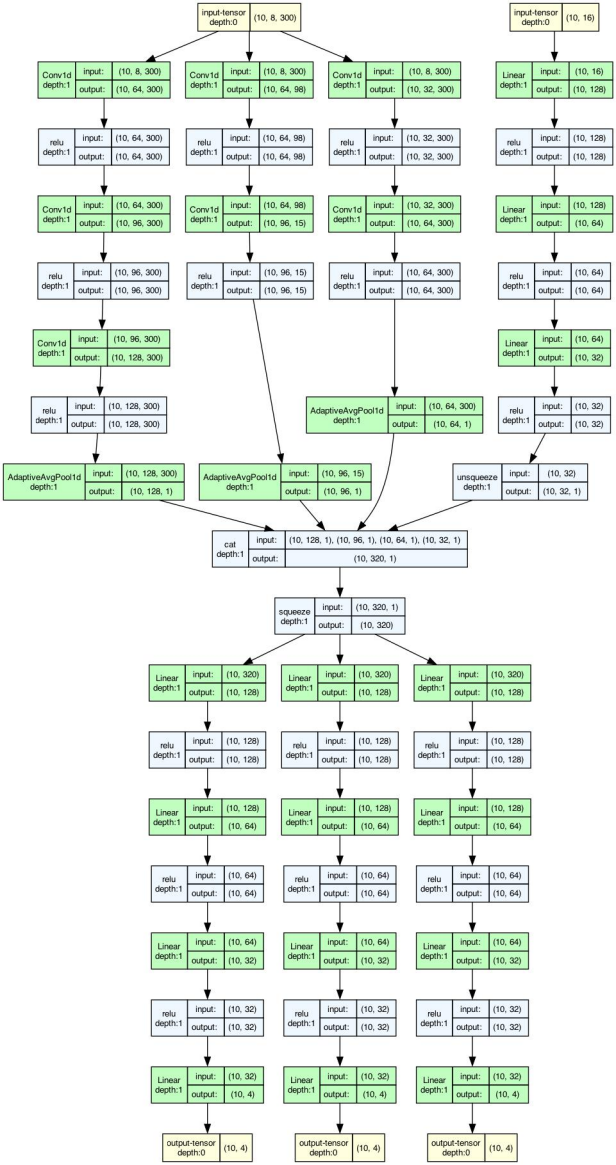

Train
Validation
0.14
0.12
0.10
0.08
0.06
0.04
loss_upper
0
10
20
30
40
50
60
Epochs


Train
Validation
loss_value
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60
Epochs


Train
Validation
mae_value
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0
10
20
30
40
50
60
Epochs


Train
Validation
medape_value
Epochs
50
40
30
20
10
0
10
20
30
40
50
60


Train
Validation
mse_value
Epochs
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 4)
Linear depth:1
input: (10, 32)
output: (10, 4)
Linear depth:1
input: (10, 32)
output: (10, 4)
output-tensor depth:0
(10, 4)
output-tensor depth:0
(10, 4)
output-tensor depth:0
(10, 4)